

Quiz — 2.2.1 Programming Techniques

Name: _____ Date: _ **Mark:** ____ / 25

OCR H446 · Component 02 · 2.2.1

Section A — Multiple choice (8 marks)

A1. Which loop is guaranteed to execute its body **at least once**? A. `while` B. `for...next` C. `do...until` D. `if...else` Your answer: ☐

A2. What is the defining difference between a function and a procedure? A. A function can take parameters; a procedure cannot B. A function returns a single value; a procedure does not return a value C. A procedure is recursive; a function is iterative D. A function is global; a procedure is local Your answer: ☐

A3. A recursive subroutine with no reachable base case will most likely cause: A. A syntax error B. An infinite `while` loop C. A stack overflow D. A logic error that returns 0 Your answer: ☐

A4. When an argument is passed **by reference**, what is passed to the subroutine? A. A copy of the value B. The memory address of the original variable C. The data type only D. A new global variable Your answer: ☐

A5. A variable declared inside a subroutine is: A. Global, visible everywhere B. Local, visible only within that subroutine C. A constant D. Automatically passed by reference Your answer: ☐

A6. Which OOP concept keeps attributes **private** and provides public methods to access them? A. Inheritance B. Polymorphism C. Encapsulation D. Overloading Your answer: ☐

A7. A subclass replaces an inherited method with one of the **same name and parameters**. This is: A. Overloading B. Overriding C. Instantiation D. Encapsulation Your answer: ☐

A8. Which IDE feature lets a programmer **pause execution at a chosen line** to inspect variables? A. Syntax highlighting B. Auto-indentation C. Breakpoint D. Code completion Your answer: ☐

Section B — Short answer (12 marks)

B1. State the **two** parts every correctly written recursive subroutine must contain, and give the role of each. [2]

B2. Consider the program below. State the value printed and explain why, referring to **scope**. [3]

```
total = 10

procedure update()
    total = 5
    print(total)
endprocedure

update()
print(total)
```

Note: in this language an assignment to `total` inside the procedure creates a **new local** variable `total`; it does not change the global.

Output line 1: __ **Output line 2:** ____

Explanation: _____

B3. A subroutine is called with `x = 3`. Explain what the **final value of x** is after each call below, and why. **[3]**

```
procedure addOneByVal(byVal n)
    n = n + 1
endprocedure

procedure addOneByRef(byRef n)
    n = n + 1
endprocedure
```

After `addOneByVal(x)` : `x =` __ **because** _____

After `addOneByRef(x)` : `x =` __ **because** _____

B4. Trace the call `factorial(4)` for the routine below. Show the value returned at each level and the final result. **[2]**

```
function factorial(n)
  if n <= 1 then
    return 1
  else
    return n * factorial(n - 1)
  endif
endfunction
```

B5. Explain **one** advantage of preferring **local** variables over global variables. **[1]**

B6. Define the terms **class** and **object**, making clear how they relate. **[1]**

Section C — Exam-style question (5 marks)

A `BankAccount` class is being written. Complete the OCR-style class so that it:

- has a **private** attribute `balance` initialised to 0 by the constructor,
- has a method `deposit(amount)` that adds `amount` to the balance,
- has a method `getBalance()` that returns the current balance.

```
class BankAccount
  private balance

  public procedure new()
    // (a) initialise balance

  endprocedure

  // (b) deposit method

  // (c) getBalance method

endclass
```

Write your answers for (a), (b) and (c) below. **[5]**
